

MACHINE LEARNING (ML) — Complete Roadmap & Interview Index

B.Tech CSE 4th Year • Basic → Intermediate → Advanced • Interview + Online Assessment Preparation

How to use this index: Follow the numbered sequence. Each topic is intentionally listed as a future chapter/lesson rather than explained here. The roadmap covers learning order, practical skills, interview depth, coding/OA relevance, projects, deployment, and revision.

A. Common Prerequisites & Interview Core

Purpose: Shared foundation before specializing.

A.1 Programming Fundamentals — Variables, data types, operators, input/output, conditions, loops, functions, scope and recursion.

A.2 Python for Technical Interviews — Lists, tuples, sets, dictionaries, strings, slicing, comprehensions, functions, modules and exceptions.

A.3 OOP Essentials — Classes, objects, inheritance, polymorphism, abstraction, encapsulation and composition.

A.4 DSA Foundation — Arrays, strings, linked lists, stacks, queues, hash tables, trees, heaps, graphs and recursion.

A.5 Algorithms — Sorting, searching, two pointers, sliding window, prefix sums, greedy, divide-and-conquer and dynamic programming.

A.6 Complexity — Big-O, time/space complexity, amortized thinking and trade-offs.

A.7 SQL Basics — SELECT, filtering, joins, grouping, subqueries, CTEs, window functions and query optimization basics.

A.8 Git & GitHub — Repositories, branching, commits, merge/rebase, pull requests and collaborative workflow.

A.9 Linux & CLI — Files, permissions, processes, pipes, grep, sed, awk, package management and environment variables.

A.10 Interview Communication — Explain approach, assumptions, complexity, trade-offs, debugging and project decisions.

1. ML Foundations

Purpose: Build the conceptual base before algorithms.

1.1 What is Machine Learning? — Learning patterns from data instead of explicitly programming every rule.

1.2 Learning Types — Supervised, unsupervised, semi-supervised and reinforcement learning.

1.3 Problem Framing — Features, targets, training objective, constraints and success metrics.

1.4 Dataset Fundamentals — Rows, columns, labels, distributions, missing values and outliers.

1.5 Train/Validation/Test — Correct splitting, stratification, temporal splits and leakage prevention.

1.6 ML Workflow — Problem → data → EDA → preprocessing → baseline → model → validation → tuning → deployment.

2. Mathematics & Statistics

Purpose: The quantitative foundation behind model behavior.

2.1 Linear Algebra — Vectors, matrices, dot products, norms, projections and eigen concepts.

2.2 Probability — Conditional probability, Bayes theorem, distributions and expectation.

2.3 Statistics — Descriptive statistics, sampling, confidence intervals and hypothesis testing.

2.4 Correlation & Covariance — Relationships between variables and their limitations.

2.5 Calculus — Derivatives, gradients, chain rule and optimization.

2.6 Optimization — Objective functions, gradient descent, learning rates and convergence intuition.

3. Data Preparation

Purpose: Make raw data suitable for reliable learning.

3.1 Data Cleaning — Missing values, duplicates, inconsistent formats and noisy records.

3.2 Outlier Handling — Detection, robust statistics, transformation and domain-aware treatment.

3.3 Encoding — Label, ordinal, one-hot, target and frequency encoding.

3.4 Scaling — Standardization, normalization and when scaling matters.

3.5 Feature Engineering — Transforms, interactions, aggregations, time features and domain features.

3.6 Feature Selection — Filter, wrapper and embedded approaches.

3.7 Pipelines — Reusable preprocessing/model pipelines and leakage-safe transformations.

4. Supervised Learning

Purpose: Core regression and classification algorithms.

4.1 Linear Regression — OLS intuition, assumptions, coefficients and residuals.

4.2 Regularized Regression — Ridge, Lasso and Elastic Net.

4.3 Logistic Regression — Probability, sigmoid, decision boundary and regularization.

4.4 k-NN — Distance metrics, choosing k and scaling.

4.5 Naive Bayes — Conditional independence and probabilistic classification.

4.6 Decision Trees — Splits, impurity, information gain and pruning.

4.7 SVM — Margins, kernels and classification intuition.

4.8 Ensemble Learning — Bagging, random forests, boosting and gradient boosting.

4.9 XGBoost/LightGBM Concepts — Tree boosting, regularization, learning rate and practical tuning.

5. Unsupervised Learning

Purpose: Find structure without target labels.

5.1 Clustering — K-means, hierarchical clustering and DBSCAN.

5.2 Cluster Evaluation — Silhouette score, inertia and domain validation.

5.3 Dimensionality Reduction — PCA and variance-preserving projection.

5.4 Anomaly Detection — Statistical, distance-based and isolation-based approaches.

5.5 Association Rules — Support, confidence, lift and market-basket use cases.

6. Model Evaluation & Tuning

Purpose: Learn to select models that generalize.

6.1 Regression Metrics — MAE, MSE, RMSE, R^2 and MAPE caveats.

6.2 Classification Metrics — Accuracy, precision, recall, F1, ROC-AUC and PR-AUC.

6.3 Confusion Matrix — TP, TN, FP, FN and business interpretation.

6.4 Cross-Validation — K-fold, stratified, grouped and time-series validation.

6.5 Hyperparameter Tuning — Grid search, random search and Bayesian optimization concepts.

6.6 Bias-Variance — Underfitting, overfitting and generalization.

6.7 Calibration — Probability quality, threshold tuning and cost-sensitive decisions.

7. Advanced ML

Purpose: Topics expected for stronger ML interviews.

7.1 Feature Engineering at Scale — Sparse features, high cardinality and efficient transformations.

7.2 Imbalanced Learning — Class weights, resampling, SMOTE concepts and correct metrics.

7.3 Time-Series ML — Trend, seasonality, lag features, rolling windows and temporal validation.

7.4 Recommenders — Collaborative filtering, content-based methods and ranking concepts.

7.5 Explainability — Feature importance, permutation importance, SHAP/LIME concepts.

7.6 Causal Thinking — Correlation vs causation, confounding and intervention intuition.

7.7 Model Robustness — Distribution shift, adversarial considerations and uncertainty.

8. ML Engineering & MLOps

Purpose: Productionize models rather than stopping at notebooks.

8.1 Scikit-learn Workflow — Estimator API, pipelines, persistence and reproducibility.

8.2 Experiment Tracking — Parameters, metrics, artifacts and model comparison.

8.3 Model Serving — Batch vs real-time inference and REST APIs.

8.4 Docker — Images, containers, dependencies and reproducible deployment.

8.5 Monitoring — Data drift, concept drift, performance and system metrics.

8.6 CI/CD for ML — Testing, validation, packaging and deployment pipelines.

8.7 Model Lifecycle — Versioning, registry, rollback and retraining.

9. ML Projects & Interview Mastery

Purpose: Demonstrate practical competence.

9.1 End-to-End Project — Business problem → data → EDA → baseline → model → tuning → evaluation.

9.2 Deployment Project — Serve a trained model through API/UI and document it.

9.3 ML System Case Study — Design data flow, training, inference, monitoring and retraining.

9.4 Interview Questions — Algorithm comparisons, assumptions, metrics, failure modes and trade-offs.

9.5 OA Preparation — Python, DSA, SQL, statistics, ML MCQs and timed case questions.

9.6 Mock Rounds — Coding + ML theory + project defense + behavioral.

FINAL INTERVIEW / OA FLOW

Phase 1: Fundamentals → **Phase 2:** Core concepts → **Phase 3:** Coding & problem solving → **Phase 4:** Tools/frameworks → **Phase 5:** Projects → **Phase 6:** System/design thinking → **Phase 7:** Mock OAs → **Phase 8:** Technical + HR interviews → **Phase 9:** Revision and company-specific preparation.

Recommended answer depth: Definition → intuition → syntax/formula → small example → edge cases → complexity/trade-offs → real-world use → common interview questions.

End goal: Be able to explain concepts without notes, implement core algorithms/models, debug code, interpret results, discuss project decisions, and solve timed OA questions.